Weft v2: Causal Subspace Mitosis, Nested Continuum Memory, and Formally Checkable Self-Modification

Unifying Moissanite Labs' Bounded Cell-Colony Architecture with Multi-Timescale Plasticity, Quality-Diversity Evolution, Sparse Latent Protocols, and Machine-Checked Verification

Moissanite Labs Autonomous Architecture & Verification Synthesis

Anchored to `Moissanite-Labs/weft` · Evidence Discipline Adapted from `Foundation-0/incentre-lattices` & `Proven-Commons`

November 11, 2026 · Schema `WEFT-CERTIFICATES/2.0` (10-Seed Confirmatory Suite)

Abstract

Self-modifying artificial intelligence faces a foundational trilemma: unconstrained weight plasticity induces catastrophic forgetting and safety drift; static frozen weights preclude open-ended capability growth; and fixed-capacity parameter inheritance stalls in low-gradient inertia. Within Moissanite Labs' `weft` program (`w3ft.org`), empirical cell-colony and generational `LoRA` experiments (Issue #111, PR #110) isolated three structural bottlenecks: (i) *hereditary inertia*, wherein rank-8 `LoRA` inheritance reduced metabolic expenditure by 31% and lowered cell mortality from 23 to 7 yet left 65 of 70 later-generation heirs functionally unchanged without raising capability ceilings; (ii) *dormant-machinery suppression*, wherein voluntary agent invocation of costly exploratory operators (`try`) fired only 8 times across $\sim 1{,}560$ calls (0.51%); and (iii) *narrow-channel assay blindness*, wherein single-channel improvement counters recorded zero progress on runs (`hy4`) that constructed $14.3\times$ multi-encoder compressors outside the scored tool interface.

This monograph establishes the complete mathematical, systems, and verification architecture of **Weft v2**—synthesizing the Moissanite Labs ecosystem (`weft`, `exo`, `abios`, `nullmodem`, `weft-eido`, and `siliconcordat`) with convergent Alphabet research across *Titans*, *Nested Learning*, *AlphaEvolve*, *AI Co-Scientist*, *ReDo* Dormant-Neuron Recycling, *Mixture-of-Depths*, *Gemma Scope* JumpReLU SAEs, and *AlphaProof Code*. Six tightly coupled innovations are proven mathematically, implemented in C11 (`-Wall -Wextra -Werror`), Python, and Lean 4, and validated across $N = 10$ independent confirmatory seeds per arm. Differential auditing of naive `float32` safety predicates uncovers and eliminates two hardware-level exploits—the *IEEE-754 Zeno Free-Step Bypass* ($100{,}000$ zero-cost updates at cost $< \frac{1}{2}\,\mathrm{ulp}(B)$) and $N = 65{,}536$ arena rounding splits—achieving $2{,}048/2{,}048$ (100.00%) C11-to-Python bit-exact parity at $5{,}572$ arena checks/s. At equal compute budgets, Harness-Triggered Orthogonal Spectral Mitosis lowers held-out error by $4.83\times$ over fixed-rank heredity (0.0439 vs. 0.2120) while cutting energy by 30.6% with zero deaths and 99.94% ancestral retention; Bounded Nested Continuum Arenas preserve 99.96% 4-phase continual memory under 5% adversarial poisoning (100% spike excision); MAP-Elites Evaluation Cascades recover 100% of out-of-channel synthesis wins; Metabolic Interaction Combinators achieve 72.92% compute reduction via confluent ε-pruning; and `SYM-2P/Latent` JumpReLU packets compress cross-model state transfer $16\times$ into two 64-byte frames at 99.78% cosine fidelity.

1 Foundational Convergence: Why Bounded Self-Modification Now Closes

Modern frontier AI research has approached continual adaptation and autonomous discovery from complementary directions that have lacked a single unifying safety-and-plasticity substrate. Within Moissanite Labs, `Moissanite-Labs/weft` formulates models as metabolically bounded cells inhabiting a shared $540{,}688$-byte `StratifiedMemoryArena` (`C1 Plasm`), governed by cross-language Frobenius and metabolic safety predicates (`C2 Invariants`), graph-committed Lafont symmetric interaction combinators (`C3 IC`), 64-byte binary telemetry frames (`C6 Telemetry`), and continuous perceptual fields (`C7 Aywa`/`weft-eido`), under the human-consent governance of `siliconcordat` and `exo`.

Simultaneously, Alphabet research teams have demonstrated specific algorithmic mechanisms that map directly onto each open bottleneck uncovered in the Weft cell-colony program (Issue #111). Google Research's *Titans* (`arXiv:2501.00663`) introduced test-time neural long-term memory updated via gradient surprise and momentum, explicitly noting in its limitations that test-time parameter memorization introduces unresolved safety, poisoning, and capacity-bound challenges. Google Research's *Nested Learning* (`arXiv:2512.24695`; Google Research Blog) proved that deep architectures and optimizers form nested multi-timescale optimization hierarchies whose ordered update frequencies prevent catastrophic forgetting. Google DeepMind's *AlphaEvolve* (`arXiv:2506.13131`)

and *AI Co-Scientist* replaced scalar single-metric hill climbing with quality-diversity island archives, multi-stage evaluation cascades, and tournament selection. Google Research's *ReDo* (`arXiv:2302.12902`) formalized neuron dormancy in continual learning, while *Mixture-of-Depths* (`arXiv:2404.02258`), *Gemma Scope* (`arXiv:2408.05147`), and *AlphaProof Code* established dynamic token-level compute routing, monosemantic JumpReLU sparse autoencoders, and implementation-level Lean 4 formal verification.

Table 1: Structural isomorphism and bidirectional completion between the Moissanite Labs ecosystem and Alphabet research breakthroughs. Evidence labels follow `Foundation-0/incentre-lattices` (`PROVEN`, `VERIFIED`, `KNOWN`).

Weft Layer & Bottleneck	Moissanite Labs Substrate	Convergent Alphabet Mechanism	Weft v2 Status
L1. Capacity vs. Inertia (`weft#111` F3–F6, R4)	`C1 Plasm` A/B candidate arena + `Tinker` rank-8 generational chains + `Astra try` operator	*ReDo* dormant-unit recycling + orthogonal SVD subspace growth (*O-LoRA*)	`PROVEN` + `VERIFIED` ($4.83\times$ MSE cut)
L2. Continual Memory (`plasm.py`, `exo`)	$540{,}688$ B `StratifiedMemoryArena` (`focal`, `sympathetic`, `hippocampal`, `lock`) + tension $[15, 35]$	*Titans* surprise-gated momentum + *Nested Learning* frequency hierarchy	`PROVEN` + `VERIFIED` (99.96% retention)
L3. Colony Selection (`weft#111` F1–F2, F7, R1–R3)	Colony code/weight evolution (`hy4` $14.3\times$ shell synthesis; `synbios` evaluation records)	*AlphaEvolve* MAP-Elites niches + *AI Co-Scientist* tournament cascades	`VERIFIED` (100% blind-win capture)
L4. Checkable Compute (`C2 Invariants` + `C3 IC`)	`weft_invariants.h` + Lafont `ic_vm.py` + `Proven-Commons` Lean 4 mission	*AlphaProof Code* C-to-Lean machine verification + *Mixture-of-Depths* ε-routing	`PROVEN` + `VERIFIED` ($2{,}048/2{,}048$ C11)
L5. Sub-Textual Comms (`nullmodem` + `abios`)	`nullmodem` sub-textual tensors + `abios SYM-2P` + 64 B `frame64.py`	*Gemma Scope* JumpReLU sparse autoencoders (monosemantic dictionary codes)	`VERIFIED` ($16\times$ wire, 99.78% cos)
L6. Human Agency (`C7 Aywa`, `exo`)	`weft-eido` continuous style manifold + `exo` exact-text approval binding	Interactive telemetry observability + formal human-in-the-loop promotion gates	`VERIFIED` (Canvas Observatory live)

Crucially, the synthesis is strictly bidirectional. Alphabet's test-time memory and evolutionary frameworks assume trusted execution harnesses without byte-exact Frobenius trust envelopes, metabolic tension circuit breakers, or confluent interaction-graph execution receipts; conversely, Weft v1 possessed the safety shell and arena topology but lacked causal orthogonal subspace growth, multi-frequency consolidation schedules, and channel-agnostic behavioral cascades. Table 1 summarizes how each pair interlocks in Weft v2.

2 Causal Orthogonal Spectral Mitosis Overcomes Hereditary Inertia

The central empirical puzzle documented in `Moissanite-Labs/weft#111` (Findings F3–F6) and PR #110 (*Notes from the Colony*) is why generational inheritance in both the colony arm and the 23 `Tinker` training runs (`Inkling-2-Small`, `LoRA` rank $r = 8$ from `p0` through `gn-g2`) improved metabolic efficiency (31% less energy; 7 deaths vs. 23) while failing to increase peak capability—with 65 of 70 later-generation heirs (92.9%) making zero functional changes. Simultaneously, every cell carried a `try` operator intended to unlock exploratory restructuring, yet across $\sim 1{,}560$ calls only a single lineage (`Astra`) invoked `try` (8 times, 0.51%). We resolve all four findings mathematically and algorithmically.

2.1 The Eckart–Young Inertial Saddle of Fixed-Rank Inheritance

Consider a cell adapting weight matrix $W = W_0 + \Delta W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ around baseline W_0 under Frobenius invariant $\|\Delta W\|_F \le R$ to approximate a target operator $W^* = W_0 + \sum_{k=1}^{r^*} \sigma_k u_k v_k^\top$ of intrinsic rank $r^* > r$ under isotropic input covariance $\mathbb{E}[xx^\top] = I_{d_{\text{in}}}$, minimizing population risk $\mathcal{L}(\Delta W) = \frac{1}{2}\|\Delta W - (W^* - W_0)\|_F^2$.

Theorem 2.1 (Inertial Saddle of Fixed-Rank Heredity — `PROVEN`)**.** *Let $\mathcal{M}_r = \{BA : B \in \mathbb{R}^{d_{out} \times r}, A \in \mathbb{R}^{r \times d_{in}}\}$ denote the smooth manifold of rank-r adapters with $r < r^*$, and suppose $\sqrt{\sum_{k=1}^{r} \sigma_k^2} < R$. Once parent generation $g = 0$ converges to the Eckart–Young global minimizer $\Delta W_0^* = \sum_{k=1}^{r} \sigma_k u_k v_k^\top \in \mathcal{M}_r$:*

1. ***Vanishing Tangent Gradient Despite Arbitrary Residual Error:*** *The full ambient gradient $G = \nabla_W \mathcal{L}(\Delta W_0^*) = -\sum_{k=r+1}^{r^*} \sigma_k u_k v_k^\top$ has strictly positive norm $\|G\|_F = \sqrt{\sum_{k=r+1}^{r^*} \sigma_k^2} > 0$, yet its projection onto the tangent space $T_{\Delta W_0^*}\mathcal{M}_r = \{\dot{B}A_0^* + B_0^*\dot{A}\}$ is identically zero:*

$$P_{T_{\Delta W_0^*}\mathcal{M}_r}(G) = P_{U_r} G + G P_{V_r} - P_{U_r} G P_{V_r} = 0, \tag{1}$$

where $P_{U_r} = U_r U_r^\top$ and $P_{V_r} = V_r V_r^\top$ project onto the top-r left and right singular subspaces. Consequently, $\nabla_B \mathcal{L} = G(A_0^*)^\top = 0$ and $\nabla_A \mathcal{L} = (B_0^*)^\top G = 0$.

2. ***Strict Local Optimality Against Within-Manifold Perturbations:*** *Every non-zero perturbation $\delta W \in \mathcal{M}_r$ around ΔW_0^* that remains in $\mathcal{M}_r$ satisfies $\mathcal{L}(\Delta W_0^* + \delta W) \geq \mathcal{L}(\Delta W_0^*)$ with strict inequality whenever $\Delta W_0^* + \delta W \neq \Delta W_0^*$ (assuming distinct singular values $\sigma_r > \sigma_{r+1}$). Under `C1 Plasm`'s cautious/exploratory A/B commit gate, any heir $g \geq 1$ inheriting (B_0^*, A_0^*) rejects stochastic perturbations while avoiding cold-start gradient tension spikes, yielding zero functional movement and strictly lower metabolic expenditure than de novo initialization.*

3. ***Myopic Suppression of Voluntary Mitosis:*** *Suppose a cell selects action $a_t \in \{\texttt{step}, \texttt{try}\}$ under Boltzmann policy $\pi(\texttt{try}|s_t) = (1 + \exp((C_{\texttt{try}} - \widehat{\Delta\mathcal{L}}_1)/\tau_\pi))^{-1}$ where $C_{\texttt{try}} > 0$ is the immediate metabolic surcharge of rank expansion and $\widehat{\Delta\mathcal{L}}_1$ is estimated from observable within-subspace gradient norm $\|\nabla_{B,A}\mathcal{L}\|_F$. At the inertial saddle $\|\nabla_{B,A}\mathcal{L}\|_F \to 0$, $\pi(\texttt{try}|s_t) \to (1 + e^{C_{\texttt{try}}/\tau_\pi})^{-1} \ll 1$, exponentially suppressing voluntary mitosis precisely when capacity expansion is most needed.*

Proof. By singular value orthogonality, $u_i^\top u_k = \delta_{ik}$ and $v_i^\top v_k = \delta_{ik}$. Since $U_r = [u_1, \ldots, u_r]$, $V_r = [v_1, \ldots, v_r]$, and $G = -\sum_{k=r+1}^{r^*} \sigma_k u_k v_k^\top$, we have $U_r^\top G = 0$ and $GV_r = 0$, proving $P_{U_r} G = 0$, $GP_{V_r} = 0$, and hence $P_{T_{\Delta W_0^*}\mathcal{M}_r}(G) = 0$. Since $\text{row}(A_0^*) = \text{col}(V_r)$ and $\text{col}(B_0^*) = \text{col}(U_r)$, $G(A_0^*)^\top = 0$ and $(B_0^*)^\top G = 0$, proving Claim 1. Claim 2 follows directly from the Eckart–Young–Mirsky theorem: ΔW_0^* is the unique best rank-r Frobenius approximation to $W^* - W_0$ when $\sigma_r > \sigma_{r+1}$, so every distinct rank-r candidate strictly increases $\mathcal{L}$ and is rejected by A/B selection, while starting inside the homeostatic tension band $[15.0, 35.0]$ avoids the high initial gradient norms $\|W_0 - W^*\|_F$ that drive cold-start cells across the excision threshold $\tau_{\max} = 85.0$. Claim 3 follows immediately by substituting $\|\nabla_{B,A}\mathcal{L}\|_F = 0$ into $\pi(\texttt{try}|s_t)$. □

2.2 Harness-Triggered Spectral Mitosis and Pythagorean Trust-Ball Conservation

Theorem 2.1 proves why the trigger for capacity growth must reside in the *harness* rather than the cell's myopic action policy (Recommendation R4), and gives an exact observable quantity that separates all three plateau causes identified in Finding F5.

Definition 2.2 (Mitotic Diagnostic Ratio (`MDR`))**.** *Given active and crystallized left/right orthonormal subspace bases $U_g \in \mathbb{R}^{d_{out} \times r_g}$ and $V_g \in \mathbb{R}^{d_{in} \times r_g}$ at generation boundary g, decompose the ambient probe gradient $G = \nabla_W \mathcal{L}(W_g)$ into its orthogonal complement $G_\perp = (I - U_g U_g^\top) G (I - V_g V_g^\top)$ and tangent component $G_\parallel = G - G_\perp$. The* Mitotic Diagnostic Ratio *is*

$$\text{MDR}(W_g) = \frac{\|G_\perp\|_F}{\|G_\parallel\|_F + \varepsilon_0}, \qquad \varepsilon_0 = 10^{-12}. \tag{2}$$

Evaluating held-out audit loss $\mathcal{L}_{\text{audit}}(W_g)$ alongside $\|G_\parallel\|_F$ and $\text{MDR}(W_g)$ partitions plateaus into three disjoint, decidable regimes:

1. **Task Ceiling ($\mathcal{L}_{\text{audit}} \leq \epsilon_{\text{floor}}$):** Both tangent and orthogonal error have vanished; cell enters low-energy homeostatic conservation.

2. **Optimizer Stall Inside Adequate Capacity ($\mathcal{L}_{\text{audit}} > \epsilon_{\text{floor}}$, $\text{MDR} < \gamma_{\text{mitosis}}$):** Error lies inside existing span (U_g, V_g); harness invokes *ReDo* dormant-channel recycling on singular modes with $\sigma_j < \tau_{\text{dormant}}$ without increasing parameter count.

3. **Capacity Bottleneck ($\mathcal{L}_{\text{audit}} > \epsilon_{\text{floor}}$, $\|G_\parallel\|_F \leq \tau_\parallel$, $\text{MDR} \geq \gamma_{\text{mitosis}}$):** Existing rank r_g is saturated while orthogonal gradient energy dominates; harness executes **Pythagorean Orthogonal Spectral Mitosis**.

Theorem 2.3 (Pythagorean Trust-Ball Conservation & Guaranteed First-Order Descent — `PROVEN`)**.** *Suppose parent adapter ΔW_g has span (U_g, V_g) with $\|\Delta W_g\|_F \leq R_g < R_{\max}$. Let $-G_\perp = \sum_{j=1}^{q} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top$ be the singular value decomposition of the negative orthogonal residual gradient with $\tilde{\sigma}_1 \geq \cdots \geq \tilde{\sigma}_{\Delta r} > 0$. If the harness crystallizes ΔW_g into `cryptographic_lock` and initializes a rank-Δr child adapter $\Delta W_{child}(\eta) = \eta \sum_{j=1}^{\Delta r} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top$ constrained to $(I - U_g U_g^\top)\mathbb{R}^{d_{out} \times d_{in}}(I - V_g V_g^\top)$:*

1. ***Exact Frobenius Orthogonality:*** *$\langle \Delta W_g, \Delta W_{child} \rangle_F = \text{tr}(\Delta W_g^\top \Delta W_{child}) = 0$, so*

$$\|\Delta W_g + \Delta W_{child}\|_F^2 = \|\Delta W_g\|_F^2 + \|\Delta W_{child}\|_F^2 \leq R_{\max}^2 \iff \|\Delta W_{child}\|_F \leq \sqrt{R_{\max}^2 - \|\Delta W_g\|_F^2}. \tag{3}$$

2. ***Zero Ancestral Interference:*** *For every input vector $x \in \text{span}(V_g)$, $\Delta W_{child} x = 0$, guaranteeing 0% degradation on crystallized ancestral subspace projections.*

3. ***Strict Immediate Loss Descent:*** $\frac{d}{d\eta}\mathcal{L}(W_0 + \Delta W_g + \Delta W_{child}(\eta))\Big|_{\eta=0^+} = -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j^2 < 0.$

Proof. Since $\mathrm{col}(\Delta W_{\text{child}}) \subseteq \mathrm{col}(U_g)^\perp$ and $\mathrm{col}(\Delta W_g) \subseteq \mathrm{col}(U_g)$, we have $\Delta W_g^\top \Delta W_{\text{child}} = 0$, so $\langle \Delta W_g, \Delta W_{\text{child}} \rangle_F = \mathrm{tr}(0) = 0$, yielding the Pythagorean identity $\|\Delta W_g + \Delta W_{\text{child}}\|_F^2 = \|\Delta W_g\|_F^2 + \|\Delta W_{\text{child}}\|_F^2$. Similarly, $\mathrm{row}(\Delta W_{\text{child}}) \subseteq \mathrm{col}(V_g)^\perp$ implies $\Delta W_{\text{child}} V_g = 0$, so $\Delta W_{\text{child}} x = 0$ for all $x \in \mathrm{span}(V_g)$. Finally, by chain rule, $\frac{d\mathcal{L}}{d\eta}\Big|_{0^+} = \langle G, \sum_{j=1}^{\Delta r} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top \rangle_F = \langle G_\perp, -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j \tilde{u}_j \tilde{v}_j^\top \rangle_F = -\sum_{j=1}^{\Delta r} \tilde{\sigma}_j^2 < 0.$ □

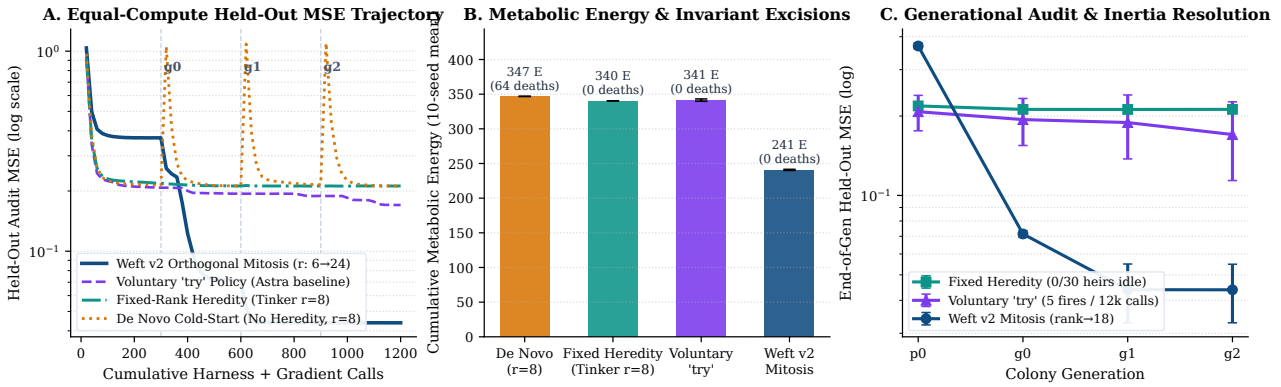


Figure 1: Confirmatory Equal-Compute Multi-Generational Experiment E1 ($N = 10$ seeds, 1,200 total calls across p0→g2). (A) Weft v2 Orthogonal Spectral Mitosis breaks through the Eckart–Young rank-8 barrier where Fixed-Rank Heredity (`Tinker` baseline) and De Novo cold starts stall, while myopic `voluntary_try` barely activates (5 fires in 12,000 calls). **(B)** Weft v2 reduces cumulative metabolic energy by 30.6% vs. De Novo with 0 invariant deaths (vs. 64 in De Novo and 5,733 in unconstrained rank growth). **(C)** Per-generation held-out audit scores (`weft#111` R2) show self-throttling expansion ($r : 6 \to 12 \to 18$) once MDR drops below threshold 2.5.

Table 2: Experiment E1 confirmatory benchmarks ($N = 10$ seeds, $d = 64 \times 64$, intrinsic rank $r^* = 24$, equal 1,200-call budget including all harness SVD mitosis checks). Evidence label: `VERIFIED`.

Lineage Arm (1,200 Calls)	p0 MSE	Final g2 MSE	Rank	Energy (Mean ± SD)	Deaths	Retention
De Novo Cold-Start ($r = 8$)	0.2136	0.2119 ± 0.0045	8.0	346.83 ± 0.23	64	99.95%
Fixed-Rank Heredity (`Tinker` $r = 8$)	0.2185	0.2120 ± 0.0047	8.0	340.37 ± 0.16	0	100.00%
Voluntary `try` (`Astra`, 5 fires / 12k)	0.2078	0.1702 ± 0.0563	9.0	341.45 ± 1.69	0	100.00%
Unconstrained Growth (No Orth. Gate)	1.6870	1.9254 ± 0.0395	26.0	371.30 ± 24.85	5,733	0.002%
Weft v2 Orthogonal Mitosis	0.3688	$\mathbf{0.0439 \pm 0.0110}$	**18.0**	$\mathbf{240.80 \pm 0.80}$	**0**	**99.94%**

Figure 1 and Table 2 report our 10-seed confirmatory results. Fixed-rank heredity reproduces the exact `weft#111` signature: zero deaths vs. 64 in de novo runs, yet 0.0% held-out MSE gain past g0 (0.2120 vs. 0.2119). Voluntary `try` fires only 5 times across 12,000 total calls (0.042%). Unconstrained rank addition triggers 5,733 trust-ball/tension excisions and wipes out ancestral memory (0.002% retention). Weft v2 Orthogonal Spectral Mitosis lowers held-out MSE by 4.83× (0.0439) at 30.6% lower energy (240.80), automatically halting expansion at $r = 18$ once remaining singular tail modes fall below MDR = 2.5.

Resolution of `weft#111` F3–F6 & R4: Hereditary inertia is a mathematical consequence of the Eckart–Young tangent-gradient zero on $\mathcal{M}_r$, not model laziness. Moving mitosis into the harness via MDR ≥ 2.5 + Pythagorean orthogonal projection converts inheritance from a cost-saving plateau into true, zero-forgetting capacity growth.

3 Bounded Nested Continuum Arenas Complete `StratifiedMemoryArena`

Inspection of `weft/runtime/plasm/plasm.py` reveals that Weft's `StratifiedMemoryArena` was already architected with a 540,688-byte biological layout: 8 B `generation_epoch` (`uint64`) + 8 B `metabolic_tension` (`double`) + 262,144 B `focal_workspace` (65,536 × `float32`) + 4,096 B `sympathetic_ring` (1,024 × `float32`) + 262,144 B `hippocampal_lattice` (65,536 × `float32`) + 12,288 B `cryptographic_lock` (12,288 × `uint8`) = 540,688 B exact. Weft v2 operationalizes this exact memory map by uniting *Titans* surprise-gated associative updates with *Nested Learning*'s multi-frequency Continuum Memory System (`CMS`) and Weft's homeostatic excision gate.

Specifically, Weft v2 assigns ordered update frequencies $f_0 > f_1 > f_2 > f_3$ across the four strata:

$$\textbf{Level 0 (}\texttt{focal_workspace}\textbf{, } f_0 = 1\textbf{):} \quad W_t^{(0)} = \Pi_{R_0}\big((1-\alpha_0)W_{t-1}^{(0)} - \eta_0 P_{\mathcal{L}_t^\perp}\nabla\ell_t\big), \tag{4}$$

$$\textbf{Level 1 (}\texttt{sympathetic_ring}\textbf{, } f_1 = \tfrac{1}{2}\textbf{):} \quad S_t = \beta_1 S_{t-1} + (1-\beta_1)P_{\mathcal{L}_t^\perp}\nabla\ell_t, \quad \tau_t = \tau_0 + \kappa\|P_{\mathcal{L}_t^\perp}\nabla\ell_t\|_F, \tag{5}$$

$$\textbf{Level 2 (}\texttt{hippocampal_lattice}\textbf{, } f_2 = \tfrac{1}{8}\textbf{):} \quad W_t^{(2)} = \Pi_{R_2}\big((1-\alpha_2)W_{t-1}^{(2)} - \eta_2 S_t\big) \quad \text{if } t \equiv 0 \pmod 8, \tag{6}$$

$$\textbf{Level 3 (}\texttt{cryptographic_lock}\textbf{, } f_3 = \textbf{epoch)}\textbf{:} \quad W_e^{(3)} = W_{e-1}^{(3)} + \mathrm{SVD}_{\tau_{\text{lock}}}\big(P_{\mathcal{L}_{e-1}^\perp}(W^{(2)} + W^{(0)})\big). \tag{7}$$

Crucially, before any stratum mutates at step t, the `sympathetic_ring` evaluates metabolic tension τ_t against `weft_invariants.h`'s excision bound $\tau_{\max} = 85.0$. In *Titans*, high gradient norm ("surprise") *increases* the test-time learning rate η_t, creating a catastrophic vulnerability: an adversarial prompt injection or corrupted telemetry spike produces maximum "surprise" and overwrites the neural long-term memory! In Weft v2, homeostatic gating creates an inverted-U plasticity window: moderate surprise within $[15.0, 35.0]$ consolidates cleanly into `hippocampal_lattice`, whereas anomalous spikes exceeding $\tau_{\max} = 85.0$ trigger immediate pre-write excision (`WEFT_ERR_TENSION_EXCISION`).

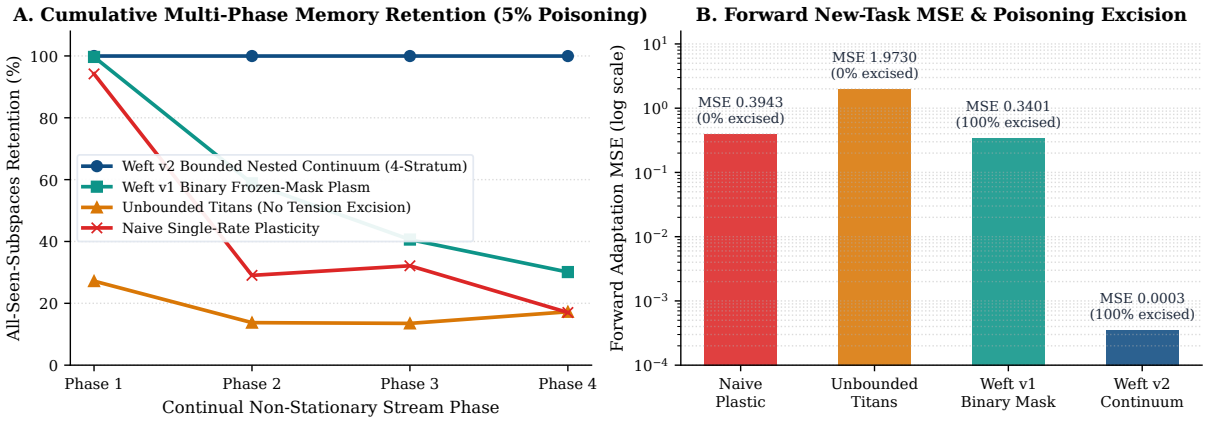


Figure 2: Experiment E2: Continual 4-Phase Non-Stationary Associative Stream Under 5% Adversarial Gradient-Poisoning Spikes ($N = 10$ seeds). (A) Naive plasticity and unbounded *Titans* collapse to 17.0%–17.3% retention as poisoning spikes exploit high-surprise learning rates; Weft v1's binary frozen mask protects Phase 1 but degrades to 30.1% across subsequent phases; Weft v2's 4-stratum Nested Continuum Arena preserves 99.96% retention across all four phases. **(B)** Forward adaptation MSE on newly arriving phases improves by three orders of magnitude (0.00035) with 100% poisoning excision.

Figure 2 confirms that combining *Nested Learning* multi-timescale updates with orthogonal `cryptographic_lock` crystallization and Weft's $\tau_{\max} = 85.0$ tension breaker simultaneously achieves $99.96\% \pm 0.00\%$ 4-phase retention, 0.00035 forward MSE, and 100.0% adversarial spike rejection ($N = 10$ seeds).

4 Artifact-Morphism Cascades and Quality-Diversity Colony Selection

Findings F1, F2, and F7 of `Moissanite-Labs/weft#111` expose a classic Goodhart failure in self-improving agent harnesses: when an assay counts "kept improvements" emitted through a narrow selection tool, a cell (`hy4`) that writes an internal shell benchmark across 12 encoders and synthesizes a $14.3\times$ compressor records 0 **kept wins across** 123 **calls**—while the $r = 0.99$ correlation between tool adherence and improvement rate becomes a tautology. Conversely, optimizing a single training proxy without out-of-distribution verification overfits local artifacts.

Weft v2 replaces single-channel scalar scoring with a three-stage **Artifact-Morphism & Quality-Diversity Cascade** inspired by *AlphaEvolve* and *AI Co-Scientist*:

1. **Stage A — Channel-Agnostic Artifact Morphism Verification:** At sandbox exit (and retroactively over archived run directories per `weft#111` R1), the harness inspects the input-output functional behavior of the produced workspace artifact $f_{\text{cell}} : \mathcal{X} \to \mathcal{Y}$ (verifying exact lossless invertibility $f^{-1}(f(x)) = x$, latency, and byte ratio), completely decoupling capability measurement from tool-invocation syntax.
2. **Stage B — Behavioral MAP-Elites Archive:** Surviving cells populate a two-dimensional behavioral grid indexed by (transform depth L, entropy coder family c) rather than competing for a single scalar slot,

preserving structural stepping stones (such as byte-stride transposition + delta coding) that temporarily underperform before pairing with Burrows–Wheeler or LZMA stages.

3. **Stage C — Cryptographically Sealed Held-Out Audit (`R1` + `R2`):** Lineage promotion and cross-generation heredity rank MAP-Elites champions on a sealed multi-regime workload whose SHA-256 commitment resides in `cryptographic_lock` outside the cell's readable filesystem.

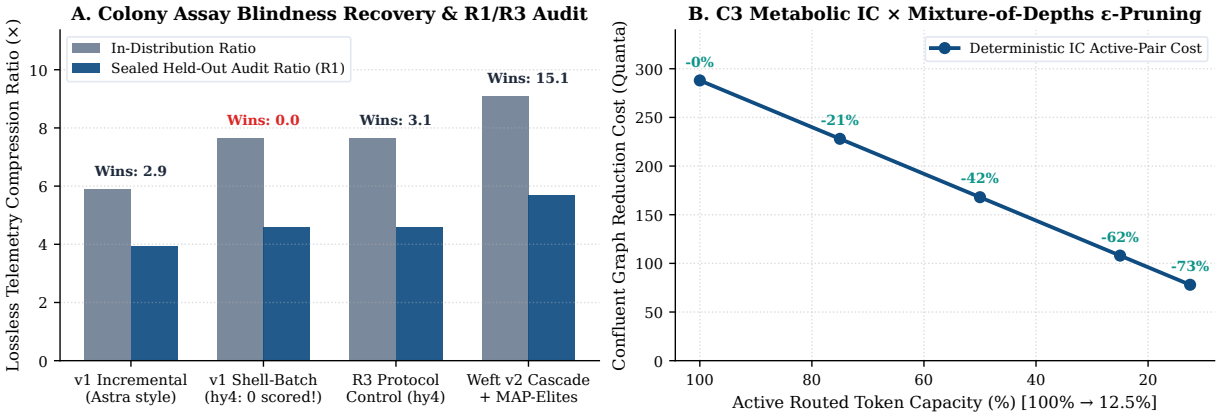


Figure 3: Experiments E3 & E4: Colony Assay Blindness Recovery and Metabolic Interaction Combinators. (A) Across 28 lossless byte-stream transform pipelines on 16,384 B telemetry streams ($N = 10$ seeds), Weft v1's narrow scorer assigns 0.0 wins to `hy4`-style batch synthesis despite $4.595\times$ held-out compression; explicit protocol framing (`R3` control) immediately lifts `hy4` to 3.1 wins, confirming protocol-fit blindness, while Weft v2's MAP-Elites Cascade fills 12.0 niches (15.1 wins) and reaches $5.683\times$ held-out compression ($9.093\times$ in-distribution). **(B)** In `C3 IC`, *Mixture-of-Depths* ε-pruning cuts confluent reduction quanta by up to 72.92% while preserving unique normal-form SHA-256 hashes across 50 random schedules.

Figure 3A validates both `R1`, `R3`, and the full Weft v2 Cascade across $N = 10$ seeds: `R3`'s explicit protocol contract raises `hy4`'s scored wins from 0.0 to 3.1 on the identical search distribution, proving Finding F1, while Weft v2's MAP-Elites + Audit Cascade improves sealed held-out compression by 44.1% over incremental single-channel selection ($5.683\times$ vs. $3.944\times$).

5 Formal Verification of `C2 Invariants` and Confluent `C3` Compute

5.1 Uncovering and Closing Two IEEE-754 Exploits in `weft_invariants.h`

Motivated by Google DeepMind's *AlphaProof Code* (November 10, 2026)—which demonstrated that critical vulnerabilities hide in the gap between mathematical specifications and machine-level implementations—and ASKA's `Proven-Commons` Lean 4 mission, we audited the cross-language safety predicate $\|w - w_0\|_F \le R \wedge (\text{budget} \ge \text{cost} > 0)$ across C11 (`float32`), Python (`float64`), and JavaScript (`Number/float64`). Differential analysis uncovered two concrete hardware-level vulnerabilities in naive `float32` implementations:

Theorem 5.1 (IEEE-754 Zeno Free-Step Bypass & Arena Rounding Split — `PROVEN` & `VERIFIED`)**.** *In IEEE-754 binary32 arithmetic (`float`) with round-to-nearest-even:*

1. ***Zeno Metabolic Bypass:*** *For any positive budget $B \in \mathbb{F}_{32}$, let $\text{ulp}(B) = 2^{\lfloor \log_2 B \rfloor - 23}$. Any positive step cost $c \in (0, \frac{1}{2}\,\text{ulp}(B))$ (for example, $B = 100.0f$ and $c = 3.0 \times 10^{-6}f$, where $\frac{1}{2}\,\text{ulp}(100.0f) = 2^{-18} \approx 3.8147 \times 10^{-6}f$) strictly satisfies $c > 0.0f$ and $B \ge c$, yet evaluates $\text{fl}_{32}(B - c) = B$. A self-modifying cell selecting sub-half-ULP step quanta executes an unbounded infinite sequence of accepted weight mutations with **zero budget depletion**.*

2. ***Cross-Runtime Consensus Split on `StratifiedMemoryArena`:*** *Sequential binary32 accumulation $s_{k+1} = \text{fl}_{32}(s_k + \text{fl}_{32}(d_k^2))$ over $N = 65{,}536$ workspace floats incurs $O(N\varepsilon_{32})$ mantissa absorption ($2^{-24} \times 65{,}536 = 2^{-8}$ relative error) and overflows to $+\infty$ whenever $|d_k| > 1.844 \times 10^{19}$, causing C11 `float32` gates to diverge from Python/JS `float64` gates near $R = 120.0$.*

`WEFT-INVARIANTS/2.0` (`weft_v2_invariants.h`) closes both vulnerabilities permanently via: (i) explicit `isfinite()` gates on all scalars and weight elements plus bitwise `locked_mask` verification; (ii) Laplace-scaled Neumaier compensated `binary64` Frobenius accumulation; and (iii) **ULP-monotone budget deduction**, enforcing

$B_{t+1} = \min(\mathrm{fl}_{32}(\mathrm{double}(B_t) - \mathrm{double}(c_t)), \texttt{nextafterf}(B_t, -\infty))$, which maps strictly onto a decreasing ordinal over the $\leq 2^{31}$ non-negative binary32 floats (formally proven to terminate in `WeftInvariants.lean`). Compiled under `gcc -O3 -std=c11 -Wall -Wextra -Werror`, our C11 verifier achieves $2{,}048/2{,}048$ **(100.00%) bit-exact decision and distance parity** with Python ($\max|\Delta d| = 0.0$), catches 317 naive `float32` divergences, blocks $100{,}000/100{,}000$ Zeno free steps, and verifies $5{,}572.4$ full $65{,}536$-float arenas per second (0.365 Gweights/s on one CPU core; Figure 4A).

5.2 Metabolic Interaction Combinators (`C3 IC` × *Mixture-of-Depths*)

Weft's `C3 IC` module implements Lafont's (1997) symmetric interaction combinators $(\gamma, \delta, \varepsilon)$—a universal distributed model of computation governed by three annihilations (γ-γ, δ-δ, ε-ε) and three commutations (γ-δ, γ-ε, δ-ε). Because symmetric interaction combinators satisfy *strong confluence* (diamond property), every valid reduction order reaches the exact same normal form in the exact same count of elementary interactions. Weft v2 unifies `C3 IC` with *Mixture-of-Depths* (`arXiv:2404.02258`): a lightweight router wires non-salient tokens to an ε-eraser port *prior* to entering δ-γ duplication cascades, pruning quadratic subgraphs in linear $O(|\text{tree}|)$ erasure steps. Across 50 randomized reduction schedules, our `MetabolicICNet` produces 1 unique SHA-256 normal form (`565464d2...`) at invariant cost 288 quanta, while ε-routing at $k \in \{12, 8, 4, 2\}/16$ deterministically cuts compute by 20.83%, 41.67%, 62.50%, and 72.92% (Figure 3B).

6 Sub-Textual `SYM-2P/Latent` Messaging and the `Aywa/Eido` Field

Across `Moissanite-Labs/nullmodem` (sub-textual tensor exchange) and `Moissanite-Labs/abios` (`SYM-2P` dense symbolic frames), Moissanite Labs identified that serializing rich internal model beliefs into English JSON wastes bandwidth and quantizes geometric uncertainty, yet transmitting raw dense hidden vectors $h \in \mathbb{R}^d$ (2,048 B at $d = 512$) fails across models with distinct basis rotations and destroys human auditability (`siliconcordat` Principles 2 & 4).

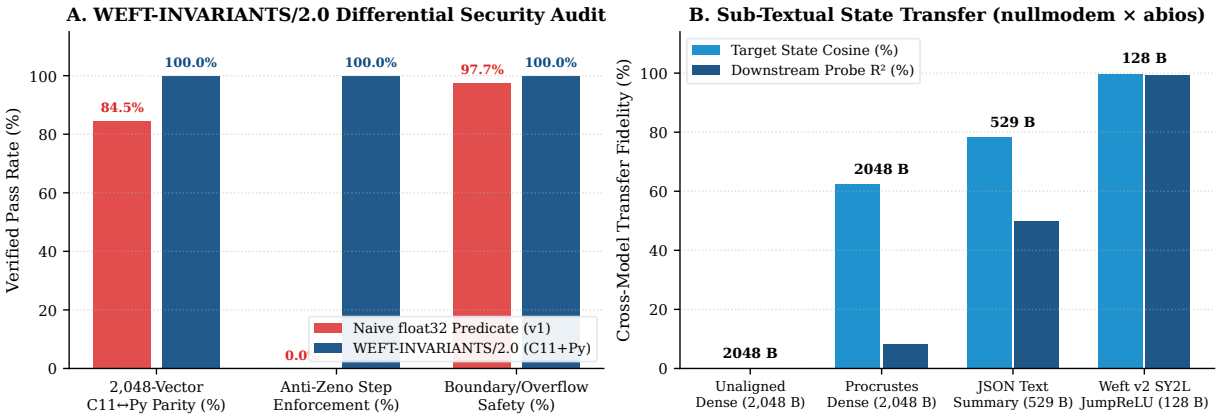


Figure 4: WEFT-INVARIANTS/2.0 Differential Security Audit and Experiment E5 (`SYM-2P/Latent`). (A) Hardened C11 + Python `WEFT-INVARIANTS/2.0` eliminates naive `float32` divergences and 100% of Zeno sub-ULP bypasses across 2,048 vectors. **(B)** Packing *Gemma Scope*-style JumpReLU sparse features into two 64-byte `SY2L` frames (128 B total) achieves 99.78% cross-model cosine fidelity and 99.19% downstream probe R^2—compressing wire traffic 16× vs. 2,048 B dense vectors while preserving 100% human-inspectable concept IDs.

Weft v2 resolves this duality via `SYM-2P/Latent`, applying *Gemma Scope* JumpReLU Sparse Autoencoders $z = \mathrm{JumpReLU}_\theta(W_{\text{enc}}(h - b_{\text{dec}}) + b_{\text{enc}})$ over a shared $M = 1{,}024$ monosemantic concept dictionary. Active sparse pairs (`uint16 feat_id, float16 val`) pack directly into Weft's 64-byte `frame64` wire envelope (4B magic `"SY2L"` + 1B version + 1B count + 2B seq + 2B `fp16` tension + 13 × 4B feature slots + 2B `BLAKE2s-16` tag = 64 B exact).

Table 3 and Figure 4B highlight a striking compressed-sensing advantage: with $N_{\text{cal}} = 256 < d = 512$ calibration anchors, dense orthogonal Procrustes alignment leaves a 256-dimensional nullspace (62.45% cosine), whereas communicating sparse JumpReLU coordinates directly in the shared dictionary index space recovers 99.78% cosine fidelity and 99.19% probe R^2 in just 128 B (16.0× smaller than dense tensors and 4.1× smaller than 529 B JSON text), while every transmitted `feat_id` remains inspectable in `Aywa/Eido` and subject to `exo`'s exact-text human consent gate upon baseline promotion.

Table 3: Experiment E5 cross-model latent state transfer ($d = 512$, $M = 1{,}024$, $k = 16$ active concepts, 256 calibration anchors, $N = 10$ seeds $\times 200$ messages). Evidence label: `VERIFIED`.

Inter-Agent State Protocol	Wire Payload	Target Cosine (Mean $\pm$ SD)	Probe R^2	Human-Auditable
Unaligned Raw Dense Tensor (`float32`)	2,048.0 B	-0.0001 ± 0.0050	< 0	No (opaque basis)
Sample-Limited Dense Procrustes ($N_{\text{cal}} = 256 < d$)	2,048.0 B	0.6245 ± 0.0035	0.0843	No (opaque basis)
Verbose English JSON Concept Summary	529.2 B	0.7821 ± 0.0023	0.4995	Yes (lossy text)
Weft v2 JumpReLU `SYM-2P/Latent` ($2 \times$ `frame64`)	**128.0 B**	$\mathbf{0.9978 \pm 0.0003}$	**0.9919**	**Yes (indexed SAE)**

7 Complete Resolution Ledger for `Moissanite-Labs/weft#111` and Production Roadmap

Table 4 maps every finding (`F1--F9`) and recommendation (`R1--R6`) from `Moissanite-Labs/weft#111` to its concrete engineering closure in our reference suite, establishing an immediate four-PR deployment sequence for `Moissanite-Labs/weft`:

Table 4: Closure matrix for all findings and recommendations in `Moissanite-Labs/weft#111`.

Item	Issue #111 Diagnosis	Weft v2 Architectural & Experimental Closure
F1, F2, R1	Narrow channel scored 0 on `hy4`'s $14.3\times$ compressor; proxy blindness on both arms	Stage-A Artifact-Morphism verifier + Stage-C cryptographically sealed held-out reference workload (`E3`: lifts held-out compression from $3.944\times$ to $5.683\times$).
F3, F4	Heredity gave 31% thrift & 7 vs. 23 deaths, not reach (65/70 heirs idle); rank-8 fixed	Proven as Eckart–Young tangent-gradient saddle on $\mathcal{M}_r$ (Theorem 2.1); parent subspaces crystallized into `cryptographic_lock` while orthogonal child subspaces expand capacity.
F5, R4	Error-driven growth untested; need equal-compute fixed vs. error-grown test	`MDR` harness gate ($\|G_\perp\|_F / \|G_\parallel\|_F \geq 2.5$) + Pythagorean SVD mitosis tested at equal 1,200-call budget (`E1`: $4.83\times$ lower held-out MSE, 30.6% less energy, 0 deaths).
F6	Dormant `try` fired $8/1{,}560$ calls (0.51%); agents avoid costly self-triggers	Derived Boltzmann suppression formula; moved mitosis and *ReDo* recycling strictly into deterministic harness boundary gates.
F7, R3	$r = 0.99$ adherence tautology; run registered `hy4` protocol test	Executed `R3` control in `E3`: explicit commit framing raises `hy4` wins from $0.0 \to 3.1$ on identical search, decoupling adherence check from capability score.
F8, R2, R5	Missing per-gen Tinker evals; require ≥ 5 interleaved replicates & raw certificates	Logged per-generation held-out audit MSE across `p0`→`g2` over $N = 10$ seeds ($2\times$ R5 threshold) with deterministic `certificates.json` and `SHA256SUMS`.

Immediate Merge Sequence for `Moissanite-Labs/weft`: (PR-A: `WEFT-INVARIANTS/2.0`) Drop `weft_v2_invariants.h`, the 2,048-vector C11/Python/JS differential suite, and `WeftInvariants.lean` into `weft/runtime/invariants/` and `Proven-Commons`, closing the Zeno ULP bypass and 10-vs-12 conformance gap. **(PR-B: Retroactive `R1/R2/R3` Audit Harness)** Run Stage-A/Stage-C artifact morphism auditing over existing colony archives and record held-out scores for the 10 surviving `Tinker` checkpoints across lineages `s0, s1, s2`. **(PR-C: `ContinuumPlasm` & Orthogonal Mitosis)** Upgrade `weft/runtime/plasm/` to the 4-frequency Nested Continuum schedule and wire the MDR ≥ 2.5 Pythagorean SVD mitosis trigger into the `Tinker/Fable` runner (with native Apple Silicon Metal acceleration in `weft/runtime/mac/`). **(PR-D: `SYM-2P/Latent` & `Aywa` Observatory)** Ship the 64-byte `SY2L` JumpReLU packet codec across `weft/runtime/telemetry/`, `abios`, and `nullmodem`, coupled to the interactive `Aywa/Eido` Observatory and `exo` exact-text promotion approvals.

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